



Fig. 2: Infrared illuminator

ambient light. As light falls on the surface of LDR1, its resistance changes. The amount of minimum light needed to actuate the relay through driver

transistor T3 can be varied by adjusting VR1. Diode 1N4001 eliminates any back voltage when the relay de-energises. Switch S1 is the mains power on/off switch and switch S2 is added to bypass the ambient light detection function.

Relay RL1 energises only when the ambient light level falls below a

threshold value set by VR1, i.e., when it's dark. Normally-opened (N/O) contacts of the relay ground path to the IR LEDs (LED1 through LED40) to make them glow. The blue LED (LED41) indicates the circuit activity. When there is ambient light and you want to use the illuminator, switch S2 'on.' All the LEDs (LED1 through LED40) glow to fulfil your requirement.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. The IR LEDs assembly is very important. A set of 40 (5x8) 5mm infra-red LEDs (IR LED1 through IR LED40) with independent current-limiting resistors (R3 through R10) per string is used. This section is powered by the input DC supply through the relay

contacts.

Mount IR LEDs on the general-purpose PCB board such that these make three circles. After soldering, carefully cut the outside of the circuit board in a round shape and fit it in a suitable metal/plastic cabinet. If available, add a suitable reflector sheet for the IR LED bank. Finally, fit the LDR bank on top of the enclosure with switches, indicator-sensitivity-control pot and power input socket. Fig. 2 shows the infrared illuminator unit.

To make the circuit actuate the relay when the intensity of ambient light is less than the preset light level, throw light on LDR1 and then slowly adjust the potentiometer until LED1 lights up and the relay energises. ●

PHOTO-COUNTER

■ RAJU R. BADDI

You can use this gadget to detect whether somebody opened your cupboard or drawer in your absence. The gadget is small and highly power-efficient. The small size means it can be easily camouflaged within boxes or containers of suitable size. The gadget is powered by a rechargeable 4.5V NiCd battery, which can be used again and again. The quiescent power consumption is extremely small, so the gadget remains active

for several days with a single charge.

Fig. 1 shows the circuit of the photo-counter. Working of the circuit is simple. When light falling on the photodiode is interrupted, counter CD4040 advances by one and the binary output is shown on the LED display. For example, if light falling on the photodiode is interrupted three times, LED1 and LED2 glow to indicate three counts.

Please note that only two output



Fig. 2: Author's prototype

connections (LED1 for LSB and LED12 for MSB) are shown in the circuit. Others are indicated through dotted lines but are not shown here. You can connect the remaining outputs of IC1 with other LEDs (LED2 through LED11, not shown here) for appropriate counting.

After assembling the circuit on a compact circuit board and giving it necessary camouflage, charge the battery once. Put the unit inside a cupboard or drawer, close the same and open it. You will find that the reading of the counter has incremented by '1' and can be read in binary fashion through LEDs. The reset button is used to reset the counter. The prototype in Fig. 2 shows the size of the gadget without LEDs.

EFY note. Try using a separate power source to power the reading LEDs. ●

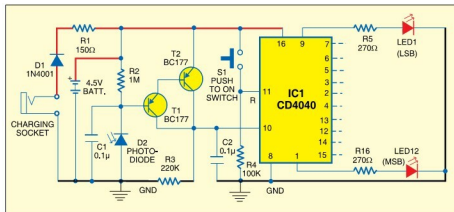


Fig. 1: Photo-counter circuit